

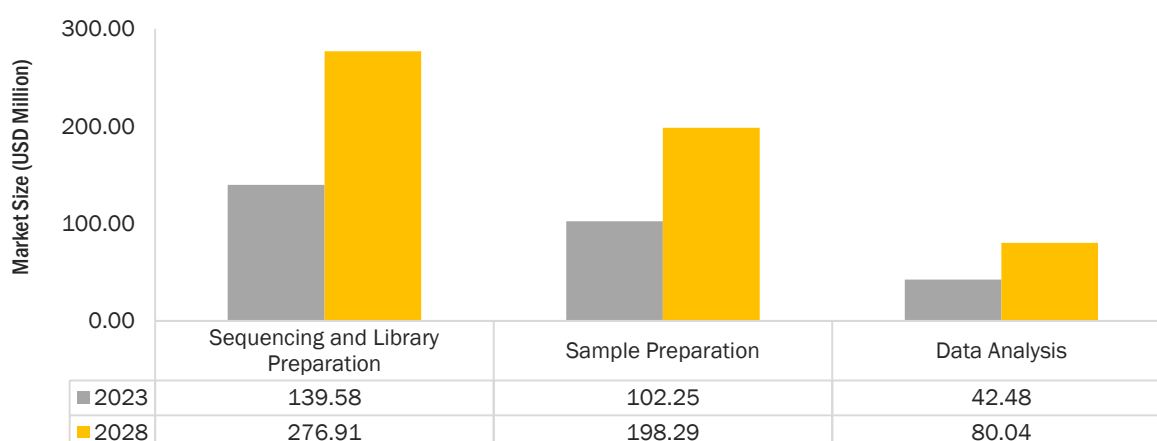
TABLE OF CONTENTS

1	INTRODUCTION.....	21
1.1	STUDY OBJECTIVES.....	21
1.2	MARKET DEFINITION.....	21
1.2.1	INCLUSIONS AND EXCLUSIONS	22
1.3	MARKET SCOPE.....	22
1.3.1	MARKETS COVERED	22
1.3.2	YEARS CONSIDERED	23
1.4	CURRENCY.....	23
1.5	LIMITATIONS	23
1.6	STAKEHOLDERS	24
1.7	SUMMARY OF CHANGES.....	24
1.7.1	RECESSION IMPACT	25
2	RESEARCH METHODOLOGY.....	26
2.1	RESEARCH DATA	26
2.1.1	SECONDARY DATA.....	27
2.1.2	PRIMARY DATA.....	28
2.2	MARKET ESTIMATION METHODOLOGY.....	29
2.2.1	INSIGHTS FROM PRIMARY EXPERTS	32
2.3	MARKET GROWTH RATE PROJECTIONS	33
2.4	MARKET BREAKDOWN AND DATA TRIANGULATION	35
2.5	RESEARCH ASSUMPTIONS	36
2.6	RISK ANALYSIS.....	36
2.7	IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET	36
3	EXECUTIVE SUMMARY.....	38
4	PREMIUM INSIGHTS	42
4.1	MICROBIOME SEQUENCING SERVICES MARKET OVERVIEW	42
4.2	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE AND COUNTRY (2023).....	43
4.3	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)	43

3 EXECUTIVE SUMMARY

The microbiome sequencing services market is projected to grow from USD 284.31 million in 2023 to USD 555.24 million by 2028, at a CAGR of 14.3%. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

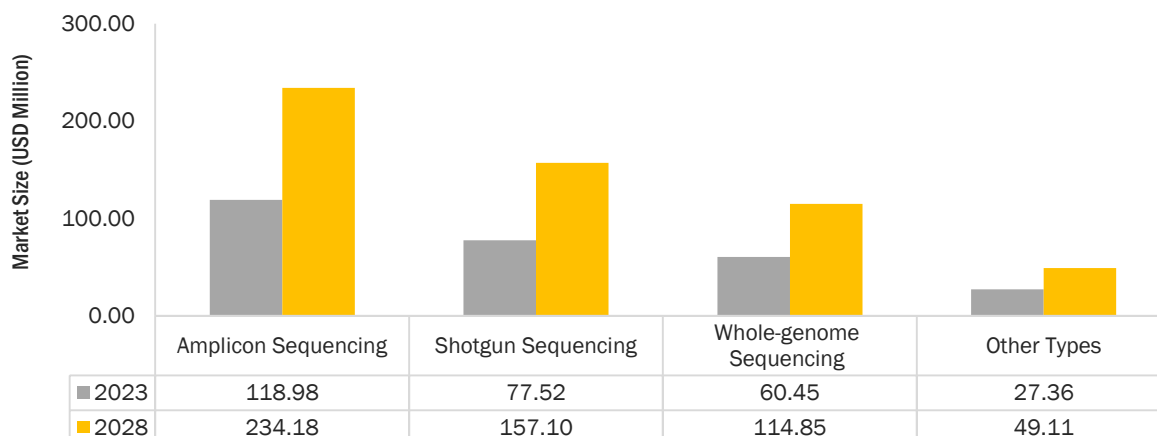
FIGURE 10 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on service, the microbiome sequencing services market is segmented into sample preparation, sequencing and library preparation, and data analysis services. A majority of the service providers/key market players offer all three types of services as part of the microbiome analysis workflow. Variations in sample size, associated pricing, use of different technologies, and the strength of local players are the key factors driving the revenue generation capacity of the above-mentioned segments. In 2022, the sequencing and library preparation segment accounted for the largest market share. This segment is also estimated to grow at the highest CAGR during the forecast period.

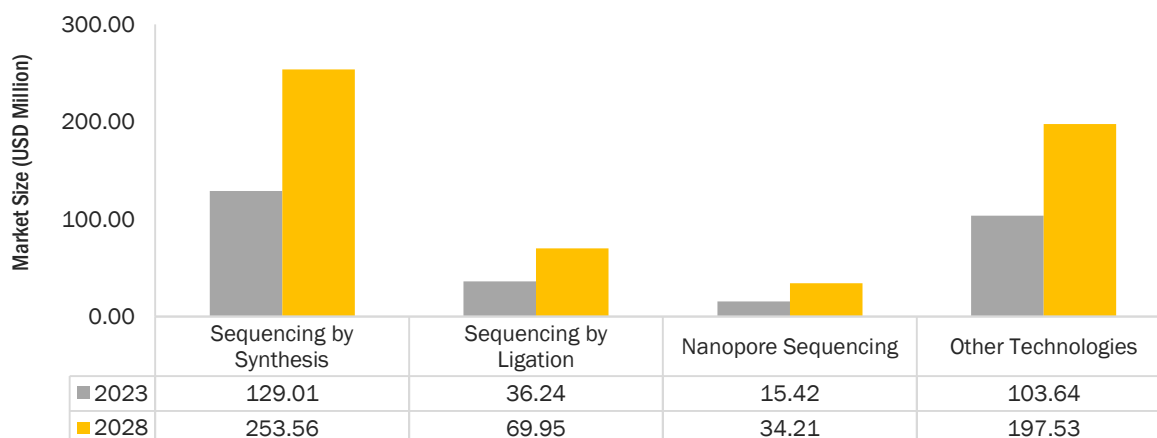
FIGURE 11 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services.

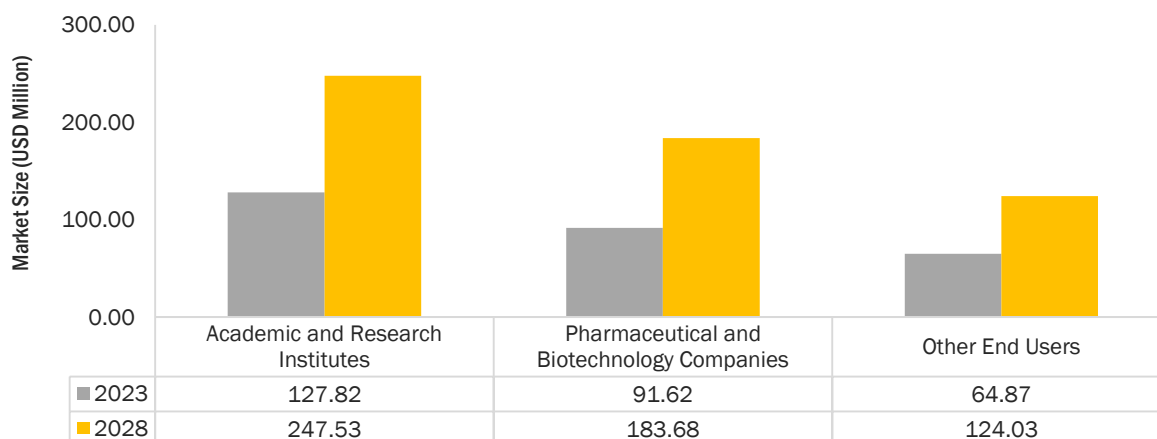
FIGURE 12 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

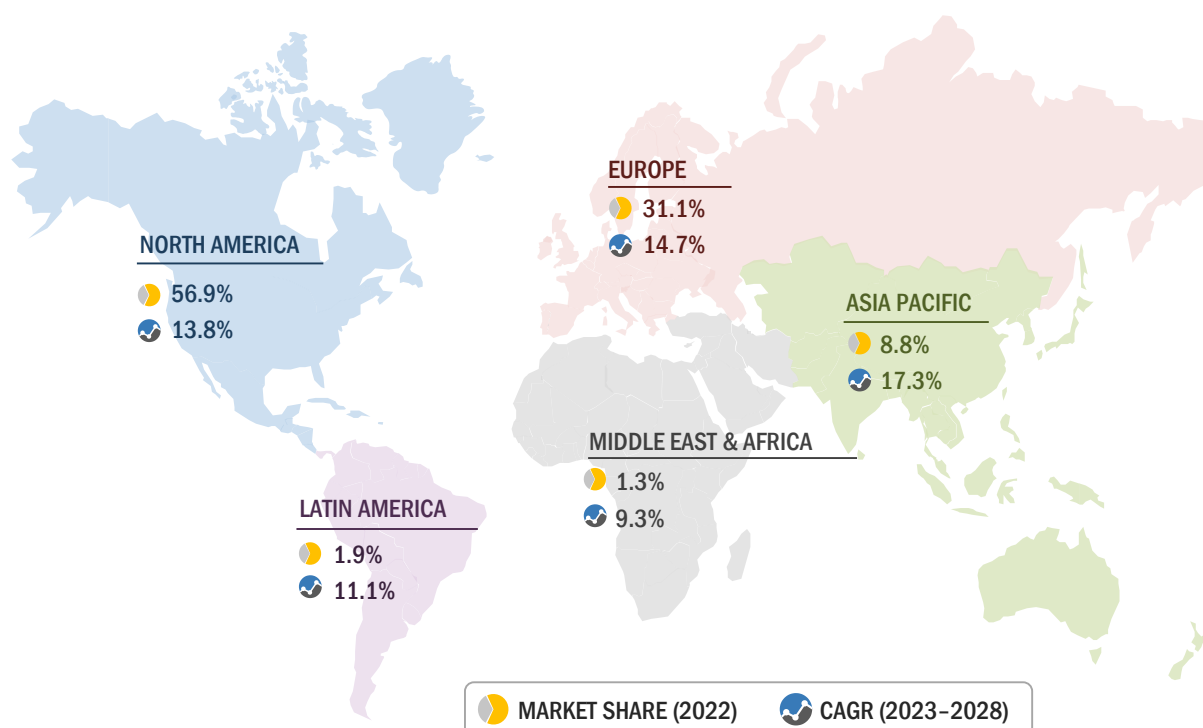
Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR during the forecast period.

FIGURE 13 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on end users, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

FIGURE 14 GEOGRAPHICAL SNAPSHOT OF MICROBIOME SEQUENCING SERVICES MARKET

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

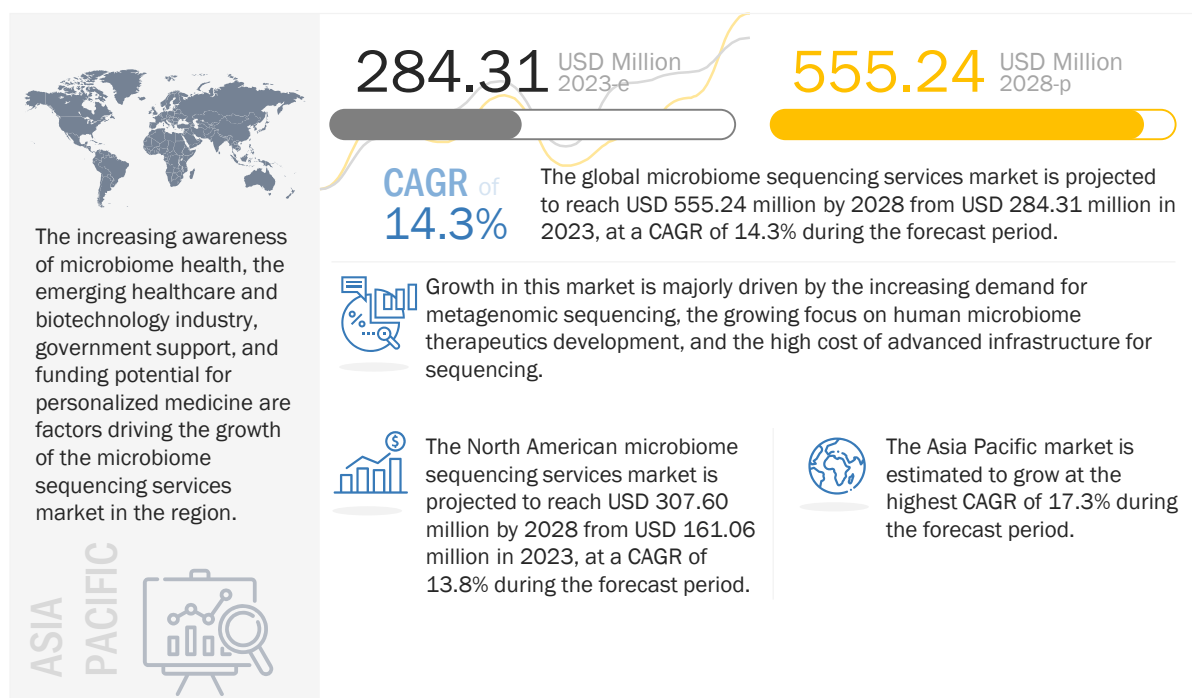
The microbiome sequencing services market is segmented into North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Factors such as established research infrastructure, technological advancements, supportive regulatory environment, robust biotechnology and healthcare industry, and the availability of funding and investment are driving the growth of the microbiome sequencing services market in North America. The microbiome sequencing services market in the Asia Pacific is expected to register the highest growth rate during the forecast period due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

The prominent players in the global microbiome sequencing services market are Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

4 PREMIUM INSIGHTS

4.1 MICROBIOME SEQUENCING SERVICES MARKET OVERVIEW

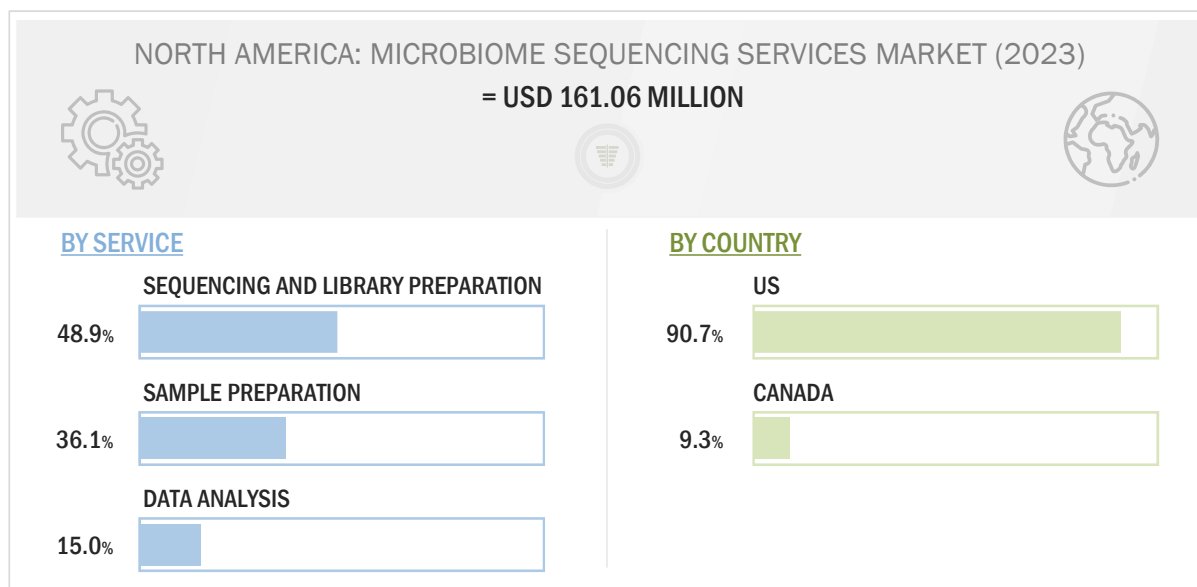
FIGURE 15 GROWING FOCUS ON HUMAN MICROBIOME THERAPEUTICS DEVELOPMENT TO DRIVE MARKET



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

4.2 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE AND COUNTRY (2023)

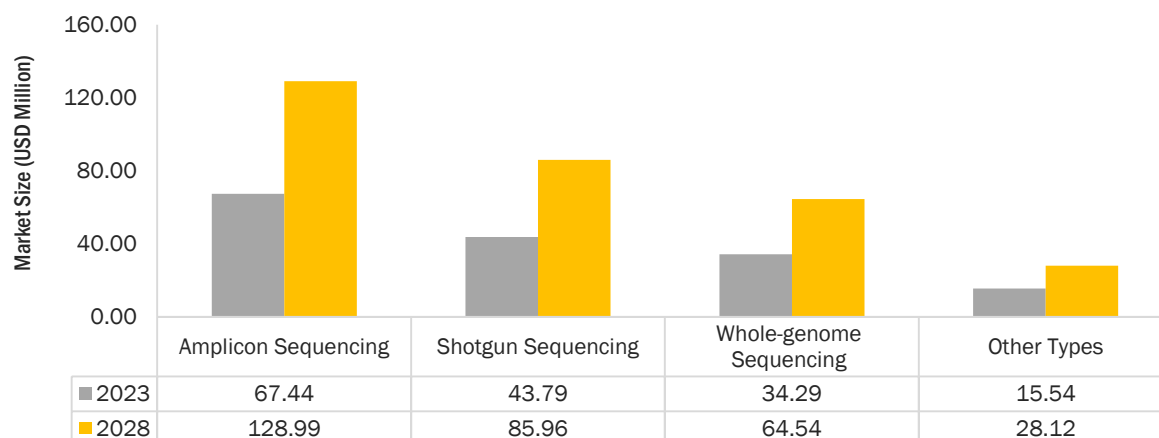
FIGURE 16 SEQUENCING AND LIBRARY PREPARATION SERVICES TO ACCOUNT FOR LARGEST SHARE OF NORTH AMERICAN MICROBIOME SEQUENCING SERVICES MARKET IN 2023



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

4.3 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)

FIGURE 17 SHOTGUN SEQUENCING SEGMENT TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis